# AN EXTENSION OF K(A,K) BY K(A,K+1) BUILT FROM THE INDEX SETS

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*Editorial note.* The counted statement is the author's own lemma environment labelled lemma-extension (source lines 3772-3794); the proof environment begins on the second line after it and is its own proof, closing at source line 3802.

*Editorial note.* Neither the statement nor the proof names another labelled result, so the card carries no context span and declares no external pointer; the definition of the Eilenberg-MacLane objects is used through the notation only.

*Editorial note.* The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

*Editorial scope.* One original result of the Stacks Project's Simplicial Methods chapter, counted once: there is a term-by-term split short exact sequence 0 to K(A,k) to E to K(A,k+1) to 0 built from the index sets.

*Allowed standard prerequisites (editorial).*

- Eilenberg-MacLane objects. The simplicial object K(A,k) with K(A,k)_n the direct sum of copies of A indexed by surjections [n] to [k], zero in degrees below k.. It is the middle term's two ends in the displayed sequence (source lines 3776-3777, 3789-3792).
- Simplicial objects defined by index sets. A simplicial object described by direct sums indexed by a set of maps out of [n], with structure maps induced by composition of indices.. It is the definition of E in the statement and what makes the maps simplicial (source lines 3778-3788, 3800-3801).
- Term by term split short exact sequences. A short exact sequence of simplicial objects which is split in every degree by the inclusions and projections of direct summands.. It is the conclusion of the statement (source line 3793).

**Lemma 0.1.** *Let $\mathcal{A}$ be an abelian category. Let $A$ be an object of $\mathcal{A}$ and let $k$ be an integer $\geq 0$. Consider the simplicial object $E$ defined by the following rules*

(1) *$E_n = \bigoplus_\alpha A$, where the sum is over $\alpha : [n] \to [k+1]$ whose image is either $[k]$ or $[k+1]$.*
(2) *Given $\varphi : [m] \to [n]$ the map $E_n \to E_m$ maps the summand corresponding to $\alpha$ via $id_A$ to the summand corresponding to $\alpha \circ \varphi$, provided $\mathrm{Im}(\alpha \circ \varphi)$ is equal to $[k]$ or $[k+1]$.*

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*Then there exists a short exact sequence*

$$0 \to K(A, k) \to E \to K(A, k+1) \to 0$$

*which is term by term split exact.*

*Proof.* The maps $K(A, k)_n \to E_n$ resp. $E_n \to K(A, k+1)_n$ are given by the inclusion of direct sums, resp. projection of direct sums which is obvious from the inclusions of index sets. It is clear that these are maps of simplicial objects. $\square$